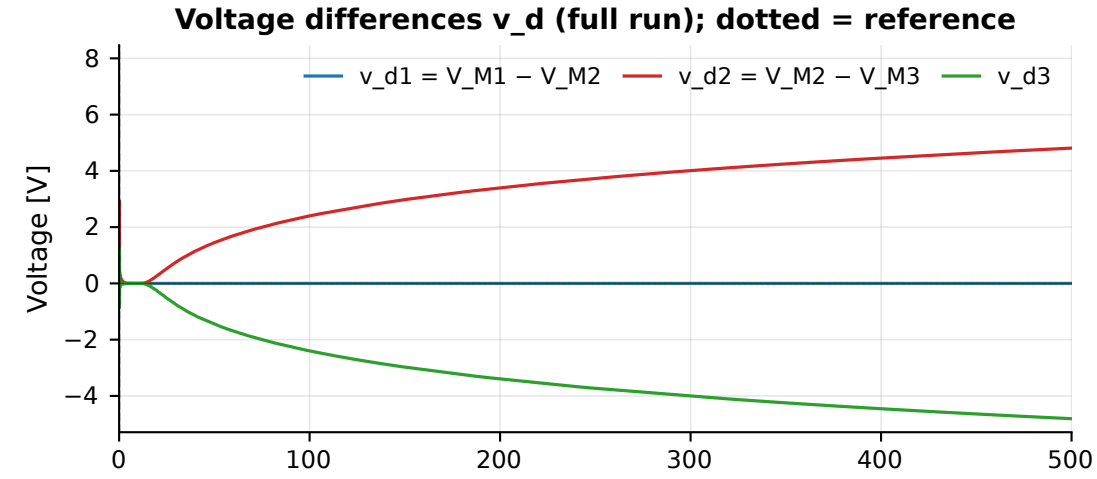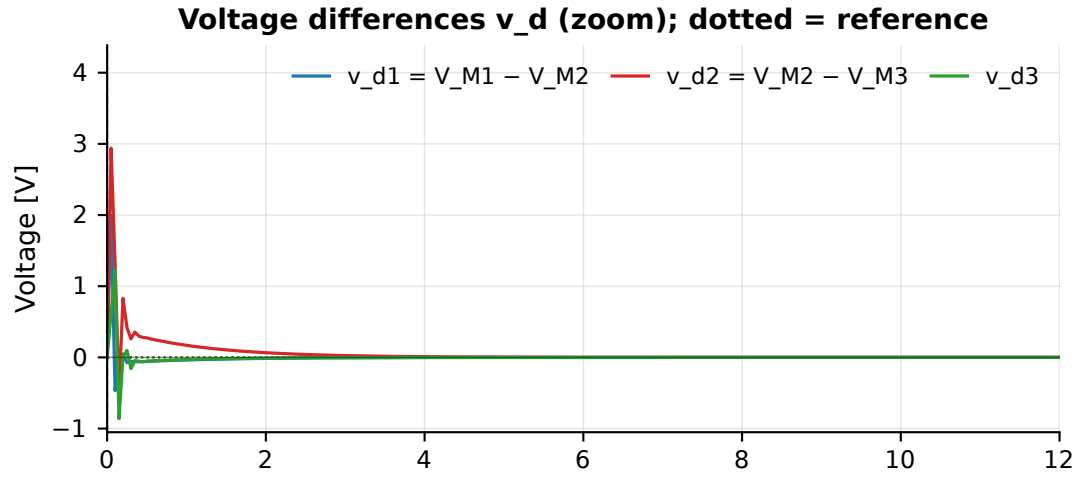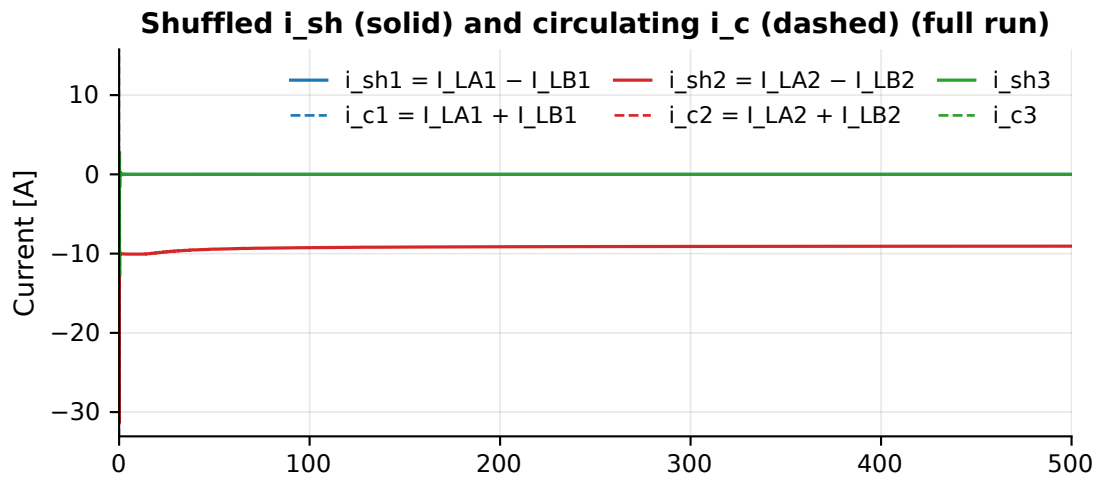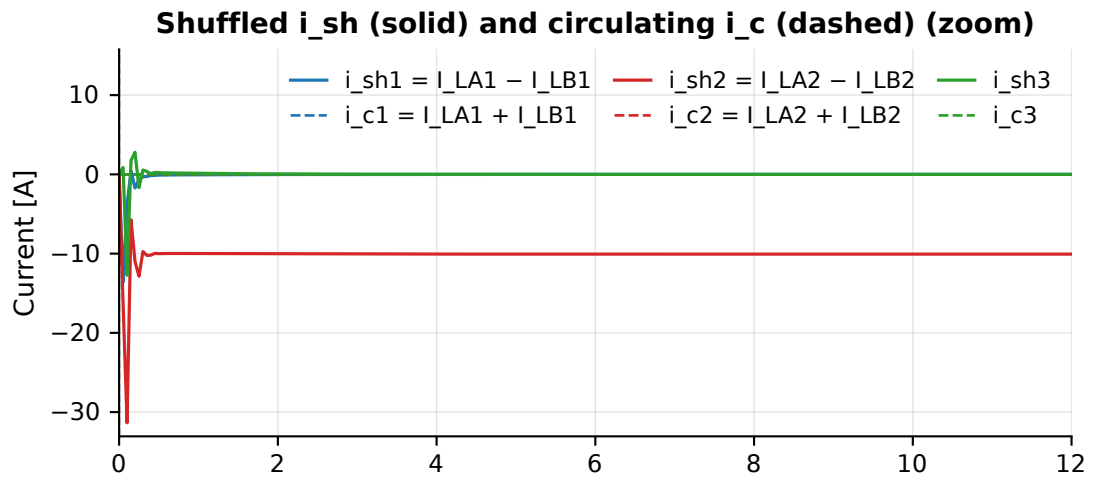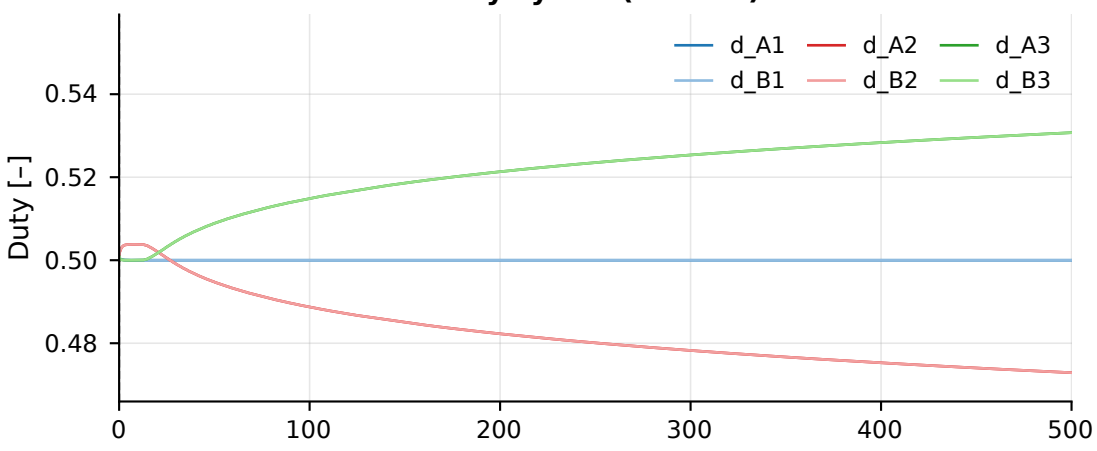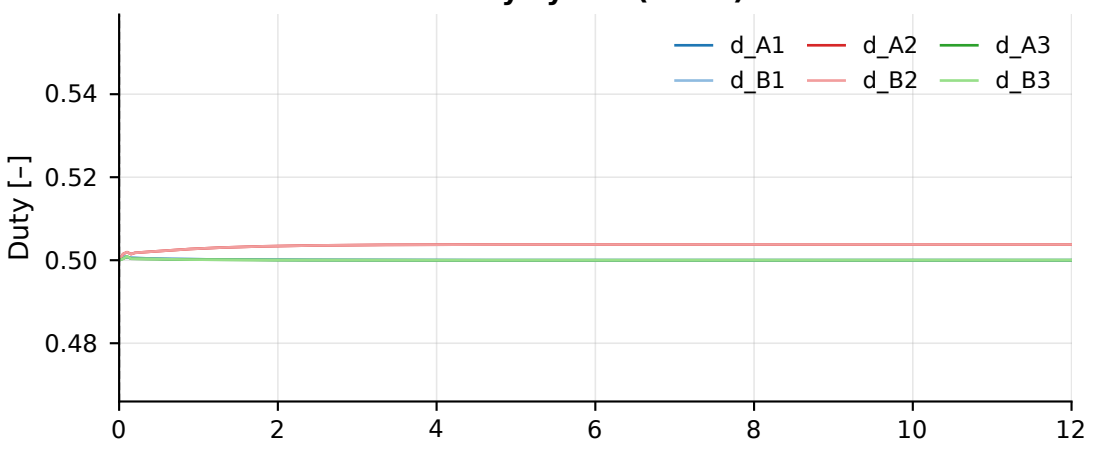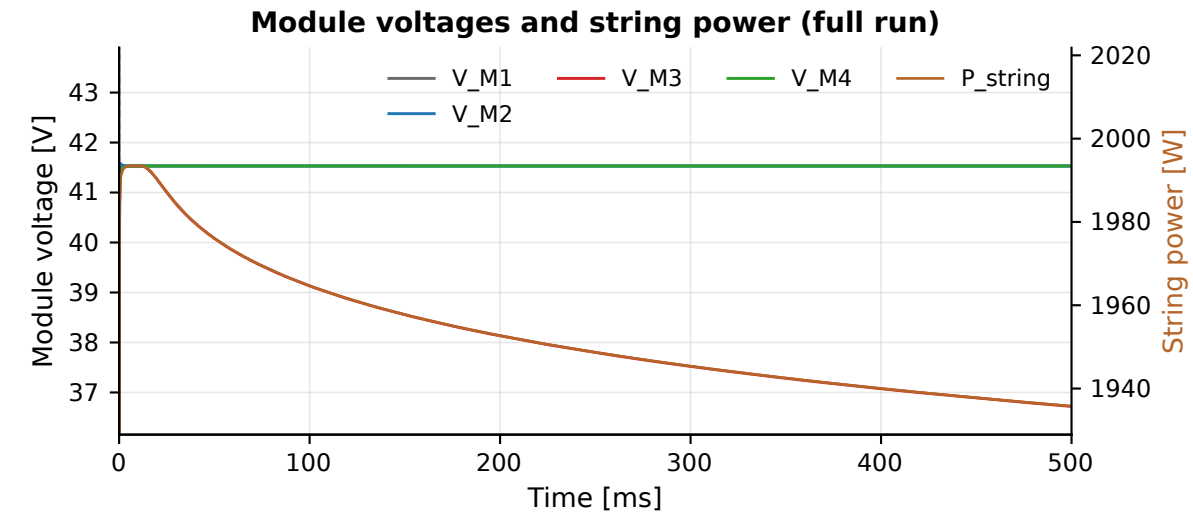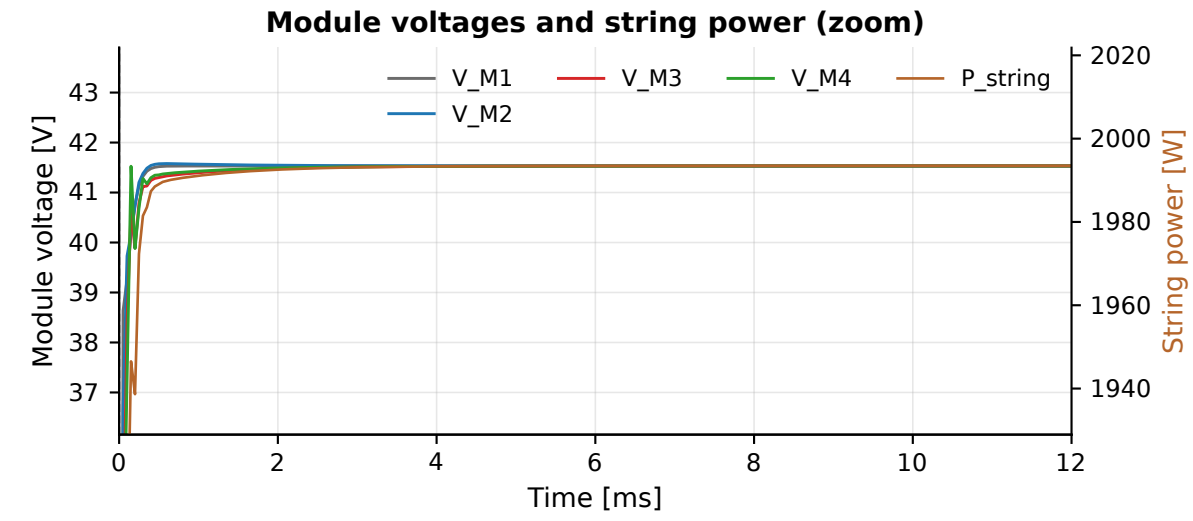MIL validate suite (4 modules): shade-deep — last module 200 W/m², refs [0, 0, 0], protect, PASS; cold start, kscale 1